



Calculating theoretical probabilities from probability tree diagrams (one event)

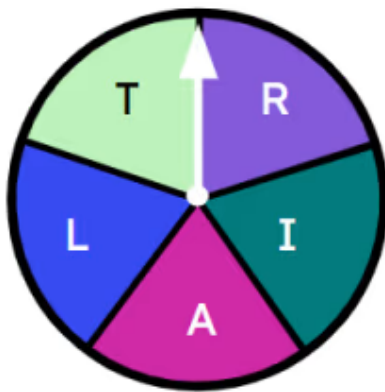
- 1 In a fair six-sided die, what is the probability of rolling a 3 and getting a head in a single toss of a fair coin? (Tick **1** correct answer)

☐ $\frac{1}{12}$

☐ $\frac{1}{6}$

☐ $\frac{1}{8}$

- 2 What is the probability that this spinner will land on R? (Tick **1** correct answer)



☐ $\frac{1}{8}$

☐ $\frac{1}{5}$

☐ $\frac{1}{3}$

- 3 A coin is flipped twice. Complete the sample space $\xi = \{HH, HT, \underline{\hspace{2cm}}, TT\}$ (Tick **1** correct answer)

☐ HH

☐ HT

☐ TH

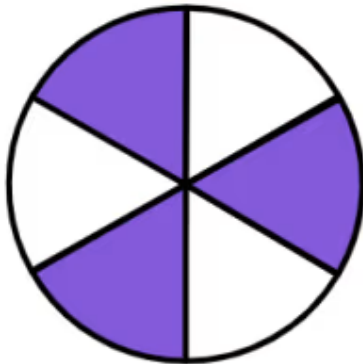
☐ TT


- 4 Andy tells his friend that this spinner is fair and that they have an equal chance of winning or losing. Is he right? (Tick **1** correct answer)



- ☐ Andy is right
- ☐ Andy is wrong
- ☐ There is no way of telling if the spinner is fair.

- 5 Sam thinks that if you place six consecutive whole numbers on each segment of this spinner, you will always have an equal chance of spinning an odd or an even number. Is Sam correct? (Tick **1** correct answer)



- ☐ Sam is correct.
- ☐ Sam is incorrect.

- 6 When rolling a fair six sided die, which event is more likely to occur? (Tick **1** correct answer)

- ☐ Rolling an even number
- ☐ Rolling a prime number
- ☐ Rolling a multiple of 1

